

# CBSE Class 12 Physics — Electric Charges and Fields

## 50 Most-Probable PYQs & Board-Pattern Practice Questions

### 1-Mark Questions (Objective/VSA)

- Q1.** An isolated point charge particle produces an electric field  $E$  at a point 3 m away from it. The distance of the point at which the field is  $E/4$  will be? [PYQ 2023] Very High
- Q2.** Two point charges placed in a medium of dielectric constant  $K = 5$  at a distance  $r$  between them experience an electrostatic force  $F$ . The electrostatic force between them in a vacuum at the same distance  $r$  will be? [PYQ 2021] Very High
- Q3.** An object has a charge of 1 C and gains  $5.0 \times 10^{18}$  electrons. What does the net charge of the object become? [PYQ 2021] High
- Q4.** What is the electric flux through a cube of side 1 cm which completely encloses an electric dipole? [PYQ 2015] Very High
- Q5. Assertion (A):** Work done in moving a charge around a closed path in an electric field is always zero. **Reason (R):** Electrostatic force is a conservative force. [PYQ 2023] Very High
- Q6.** Beams of electrons and protons move parallel to each other in the exact same direction in a vacuum. What will be the nature of the force between them? [PYQ 2023] High
- Q7.** The magnitude of the electric field due to a point charge  $2q$  at distance  $r$  is  $E$ . Then the magnitude of the electric field due to a uniformly charged thin spherical shell of radius  $R$  with total charge  $q$  at a distance  $r/2$  (where  $r \gg R$ ) will be? [PYQ 2021] Medium
- Q8. Assertion (A):** A negative charge in an electric field moves along the direction of the electric field. **Reason (R):** On a negative charge, a force acts in the direction of the electric field. [PYQ 2021] High
- Q9.** Two identical conducting balls  $A$  and  $B$  have charges  $-Q$  and  $+3Q$  respectively. They are brought in contact with each other and then separated by a distance  $d$  apart. What is the nature of the Coulomb force between them? [PYQ 2019] Very High
- Q10.** In an experiment, three microscopic latex spheres are sprayed into a chamber and become charged with charges  $+3e$ ,  $+5e$ , and  $-3e$  respectively. All three spheres come in contact simultaneously for a moment and get separated. Which of the following is a possible set of values for the final charge on the spheres: (a)  $+5e, -1e, +5e$ , (b)  $+6e, +6e, -7e$ , (c)  $-4e, +3.5e, +5.5e$ , (d)  $+5e, -8e, +7e$ ? [PYQ 2021] Medium
- Q11.** If the net electric flux through a closed Gaussian surface is zero, what can we strictly infer about the charge enclosed? [PYQ 2020] Very High
- Q12.** What is the SI unit of electric flux? [PYQ 2020] Very High
- Q13.** If the electric flux entering and leaving a closed surface in air are  $\phi_1$  and  $\phi_2$  respectively, what is the net electric charge enclosed within the surface? [PYQ 2020] High

**Q14.** An electric dipole placed in a non-uniform electric field can experience: (a) torque but not force, (b) force but not torque, (c) always a force and a torque, (d) neither a force nor a torque. [PYQ 2020] High

**Q15.** The electric field due to an infinitely long straight charged wire at a distance  $r$  is proportional to: (a)  $r$ , (b)  $1/r$ , (c)  $1/r^2$ , (d)  $1/r^3$ . [Practice] High

**Q16.** Two parallel large thin metal sheets have equal surface charge densities of  $26.4 \times 10^{-12} \text{ C/m}^2$  of opposite signs. What is the magnitude of the electric field in the region between these sheets? [PYQ 2021] Medium

**Q17.** A cylinder of radius  $R$  and length  $L$  is placed in a uniform electric field  $E$  parallel to the cylinder axis. What is the total flux for the entire surface of the cylinder? [PYQ 2022] Very High

**Q18. Assertion (A):** Two electric field lines can never cross each other. **Reason (R):** If they cross, there will be two directions of the electric field at the point of intersection, which is impossible. [Practice] Very High

**Q19.** Two charged conducting spheres of radii  $a$  and  $b$  are connected to each other by a conducting wire. What is the ratio of the electric fields at their surfaces? [PYQ 2020] High

**Q20.** A negatively charged object  $X$  is repelled by another charged object  $Y$ . However, an object  $Z$  is attracted to object  $Y$ . Which of the following is the most possible state for the object  $Z$ : (a) positively charged only, (b) negatively charged only, (c) neutral or positively charged, (d) neutral or negatively charged? [PYQ 2021] Medium

## 2-Mark Questions

**Q21.** Two identical point charges,  $q$  each, are kept  $2 \text{ m}$  apart in air. A third point charge  $Q$  of unknown magnitude and sign is placed on the line joining the charges such that the entire system remains in equilibrium. Find the position and nature of charge  $Q$ . [PYQ 2019] Very High

**Q22.** Two point charges  $+8q$  and  $-2q$  are located at  $x = 0$  and  $x = L$  respectively. Find the location of the point on the x-axis at which the net electric field due to these charges is exactly zero. [PYQ 2021] High

**Q23.** Define electric dipole moment. Is it a vector or a scalar quantity? State its SI unit. [PYQ 2019] Very High

**Q24.** Two balls of equal positive charge  $q$  Coulombs are suspended by two insulating strings of equal length from a common point. What would be the effect on the electrostatic force between them when a plastic sheet is inserted between the two balls? [PYQ 2014] High

**Q25.** Draw the pattern of electric field lines when a point charge  $-Q$  is kept near an uncharged conducting metal plate. [PYQ 2019] High

**Q26.** Depict the orientation of an electric dipole in (A) stable equilibrium, and (B) unstable equilibrium in a uniform external electric field. [PYQ 2017] Very High

**Q27.** How does the electric flux due to a point charge enclosed by a spherical Gaussian surface alter when the radius of the Gaussian surface is increased? Justify. [PYQ 2016] High

**Q28.** Two charges of magnitudes  $-2Q$  and  $+Q$  are located at points  $(a, 0)$  and  $(4a, 0)$  respectively. What is the electric flux due to these charges through a sphere of radius  $3a$  with its centre at the origin? [PYQ 2016] High

**Q29.** Calculate the exact magnitude of the electric field strength required to just support a tiny water drop of mass  $10^{-7}$  kg carrying a charge of  $1.6 \times 10^{-19}$  C from falling under gravity. (Take  $g = 9.8 \text{ m/s}^2$ ). [Practice] High

**Q30.** An electric dipole placed in a uniform electric field of intensity  $2 \times 10^5 \text{ N/C}$  at an angle of  $30^\circ$  experiences a torque equal to  $4 \text{ Nm}$ . Calculate the magnitude of charge on the dipole if the dipole length is  $2 \text{ cm}$ . [PYQ 2023] Very High

**Q31.**  $S_1$  and  $S_2$  are two hollow concentric thin spherical shells enclosing charges  $Q$  and  $2Q$  respectively. What is the ratio of the electric flux passing through  $S_1$  and  $S_2$ ? [PYQ 2014] Very High

**Q32.** A uniform electric field  $\vec{E} = 5 \times 10^3 \hat{i} \text{ N/C}$  exists in a region. Find the flux of this field through a square of side  $10 \text{ cm}$  whose plane is parallel to the y-z plane. [PYQ 2014] High

### 3-Mark Questions

**Q33.** Derive the expression for the torque acting on an electric dipole of dipole moment  $\vec{p}$  when it is held in a uniform electric field  $\vec{E}$ . Identify the orientation of the dipole in which it attains a stable equilibrium. [PYQ 2020] Very High

**Q34.** Use Gauss's law to obtain an expression for the electric field at a perpendicular distance  $r$  from an infinitely long thin straight wire with uniform linear charge density  $\lambda$ . [PYQ 2023] Very High

**Q35.** Derive an expression for the electric field intensity at any point on the equatorial line of an electric dipole. [PYQ 2019] Very High

**Q36.** Derive an expression for the electric field intensity at any point on the axial line of a short electric dipole. [Practice] High

**Q37.** Using Gauss's theorem, deduce an expression for the electric field intensity at a point near a uniformly charged infinite plane sheet of charge density  $\sigma$ . [PYQ 2017] Very High

**Q38.** A charge is distributed uniformly over a ring of radius  $a$ . Obtain an expression for the electric field intensity  $E$  at a point on the axis of the ring at a distance  $r$  from the center. [PYQ 2016] High

**Q39.** A point charge  $+q$  is placed exactly at the center of a hollow cube of side  $L$ . (A) What is the total electric flux passing through the entire cube? (B) What is the electric flux passing through any one face of the cube? (C) If the charge  $+q$  is moved to one of the corners of the cube, what will be the new total flux passing through the cube? [Practice] High

**Q40.** Two point charges of  $+1\mu\text{C}$  and  $+4\mu\text{C}$  are kept  $30 \text{ cm}$  apart. How far from the  $+1\mu\text{C}$  charge on the line joining the two charges will the net electric field be zero? [PYQ 2020] High

**Q41.** A uniformly charged large plane sheet has a surface charge density  $\sigma = \frac{1}{18\pi} \times 10^{-15} \text{ C/m}^2$ . Find the electric field at a point **A** which is **50 cm** from the sheet. Does the field change if the point is moved to **100 cm**? Why? [PYQ 2021] Medium

**Q42.** Consider two identical point charges  $+q$  located at points  $(0, 0)$  and  $(a, 0)$ . Is there a point on the line joining them at which the electric field is exactly zero? Justify your answer mathematically. [PYQ 2023] High

### 5-Mark Questions (Long Answer/Case-Study)

**Q43.** (A) State Gauss's Law in electrostatics. (B) Using Gauss's Law, derive an expression for the electric field due to a uniformly charged thin spherical shell of radius  $R$  and total charge  $Q$  at a point: (i) outside the shell ( $r > R$ ), and (ii) inside the shell ( $r < R$ ). (C) Plot a graph showing the variation of the electric field  $E$  with distance  $r$  from the center of the shell. [PYQ 2020] Very High

**Q44.** (A) Use Gauss's law to obtain an expression for the electric field due to an infinitely long thin straight wire with uniform linear charge density  $\lambda$ . (B) An infinitely long positively charged straight wire has a linear charge density  $\lambda$ . An electron is revolving in a circle with a constant speed  $v$  such that the wire passes through the center, and is perpendicular to the plane of the circle. Find the kinetic energy of the electron in terms of the magnitude of its charge and the linear charge density  $\lambda$ . [PYQ 2023] Very High

**Q45.** An electric field is uniform and acts along the  $+x$  direction in the region of positive  $x$ . It is also uniform with the same magnitude but acts in the  $-x$  direction in the region of negative  $x$ . The value of the field is  $E = 200 \text{ N/C}$  for  $x > 0$  and  $E = -200 \text{ N/C}$  for  $x < 0$ . A right circular cylinder of length **20 cm** and radius **5 cm** has its center at the origin and its axis along the  $x$ -axis so that one flat face is at  $x = +10 \text{ cm}$  and the other is at  $x = -10 \text{ cm}$ . Find: (A) The net outward flux through the entire cylinder. (B) The net charge present inside the cylinder. [PYQ 2020] Very High

**Q46.** (A) Define electric flux. Write its SI unit. (B) A hollow cylindrical box of length **1 m** and area of cross-section  **$25 \text{ cm}^2$**  is placed in a 3D coordinate system. The electric field in the region is given by  $\vec{E} = 50x\hat{i} \text{ N/C}$ , where  $x$  is in meters. Find: (i) the net flux through the cylinder, and (ii) the charge enclosed by the cylinder. [PYQ 2014] High

**Q47.** (A) Derive an expression for the electric field at any point on the equatorial line of an electric dipole. (B) Two identical point charges,  $q$  each, are kept **2 m** apart in air. A third point charge  $Q$  of unknown magnitude and sign is placed on the line joining the charges such that the system remains in equilibrium. Find the exact position, magnitude, and nature of  $Q$ . [PYQ 2019] Very High

**Q48.** (A) State Coulomb's law in vector form. (B) Three point charges  $q$ ,  $-4q$ , and  $2q$  are placed at the vertices of an equilateral triangle ABC of side  $l$ . Obtain an expression for the magnitude of the resultant electric force acting on the charge  $q$ . [PYQ 2018] High

**Q49.** (A) Derive an expression for the torque acting on an electric dipole placed in a uniform external electric field. (B) An electric dipole of length **4 cm** makes an angle of  **$60^\circ$**  with a

uniform electric field and experiences a torque of  $4\sqrt{3} \text{ Nm}$ . Calculate the magnitude of the electric field if the dipole has charges of  $\pm 8 \text{ nC}$ . [Practice] High

**Q50.** (A) Use Gauss's law to derive the expression for the electric field due to two infinite parallel plane sheets of charge having uniform surface charge densities  $+\sigma$  and  $-\sigma$ . (B) Show mathematically how the electric field behaves: (i) in the outer regions of the two plates, and (ii) in the inner space between the two plates. [Practice] Very High

## Answer Key

**Q1: 6 m.** (Since  $E \propto 1/r^2$ , to reduce the field to  $E/4$ , the distance must be doubled:  $2 \times 3\text{m} = 6\text{m}$ ). **Q2: 5F.** ( $F_{\text{vacuum}} = K \cdot F_{\text{medium}} = 5F$ ). **Q3: +0.2 C.** (Charge of electrons  $= 5.0 \times 10^{18} \times (-1.6 \times 10^{-19}) = -0.8 \text{ C}$ . Net  $= 1 - 0.8 = +0.2 \text{ C}$ ). **Q4: Zero.** (Net charge of a dipole is  $+q + (-q) = 0$ . By Gauss's Law,  $\phi = q_{\text{enclosed}}/\epsilon_0 = 0$ ). **Q5:** (a) Both A and R are true, and R is the correct explanation of A. **Q6:** Repel each other. (Currents are in opposite directions because electrons and protons have opposite charges). **Q7: 2E.** ( $E = \frac{k(2q)}{r^2} \Rightarrow k = \frac{Er^2}{2q}$ . For shell,  $E' = \frac{kq}{(r/2)^2} = \frac{4kq}{r^2} = 4(\frac{E}{2}) = 2E$ ). **Q8:** (d) Both A and R are false. (A negative charge moves opposite to the field; force acts opposite to the field). **Q9:** Repulsive. (When they touch, charges average out:  $\frac{-Q+3Q}{2} = +Q$  each. Both become positive). **Q10:** (b)  $+6e, +6e, -7e$ . (Total initial charge  $= +3e + 5e - 3e = +5e$ . Total final charge must also equal  $+5e$ . Only option (b) sums to  $+5e$ ). **Q11:** The net charge enclosed by the surface is zero ( $q_{\text{enclosed}} = 0$ ). **Q12:**  $\text{N} \cdot \text{m}^2/\text{C}$  or  $\text{V} \cdot \text{m}$ . **Q13:** Net charge  $= \epsilon_0(\phi_2 - \phi_1)$ . **Q14:** (c) always a force and a torque. (Due to varying  $E$ , forces on  $+q$  and  $-q$  do not cancel, and lines of action do not coincide). **Q15:** (b)  $1/r$ . (Since  $E = \frac{\lambda}{2\pi\epsilon_0 r}$ ). **Q16: 3 N/C.** ( $E = \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0} = \frac{\sigma}{\epsilon_0} = \frac{26.4 \times 10^{-12}}{8.85 \times 10^{-12}} = 2.98 \approx 3 \text{ N/C}$ ). **Q17:** Zero. (Flux entering circular face equals flux leaving; curved surface flux is zero as  $E$  is parallel to surface). **Q18:** (a) Both A and R are true, and R is the correct explanation of A. **Q19: b/a.** ( $V_A = V_B \Rightarrow \frac{kq_A}{a} = \frac{kq_B}{b}$ .  $E_A/E_B = \frac{kq_A/a^2}{kq_B/b^2} = \frac{q_A b^2}{q_B a^2} = \frac{ab^2}{ba^2} = \frac{b}{a}$ ). **Q20:** (d) neutral or negatively charged. (Since Z is attracted to Y, and Y repels negative X, Y is negative. Z must be positive or neutral). **Q21: 1 m** (exactly at the midpoint). Nature: Negative. Magnitude:  $Q = -q/4$ . **Q22:  $x = 2L$ .** (Equate magnitudes:  $\frac{k(8q)}{x^2} = \frac{k(2q)}{(x-L)^2}$ . Solving gives  $x = 2L$ ). **Q23:** Vector quantity.  $p = q \times 2a$ . Direction: from negative to positive charge. SI Unit: Coulomb-meter ( $\text{C} \cdot \text{m}$ ). **Q24:** Force decreases. ( $F = \frac{F_{\text{vacuum}}}{K}$ , since plastic is a dielectric with  $K > 1$ ). **Q25:** Field lines start perpendicularly from the conducting plate (which develops induced positive charge) and terminate radially inwards at the point charge  $-Q$ . **Q26:** Stable:  $\theta = 0^\circ$  ( $\vec{p}$  parallel to  $\vec{E}$ ). Unstable:  $\theta = 180^\circ$  ( $\vec{p}$  antiparallel to  $\vec{E}$ ). **Q27:** It remains unaltered. (By Gauss's law,  $\phi = q/\epsilon_0$ , which is independent of the size/radius of the Gaussian surface). **Q28:**  $-2Q/\epsilon_0$ . (Only  $-2Q$  at  $x = a$  is enclosed inside the radius  $3a$ .  $+Q$  at  $4a$  is outside). **Q29:**  $E = \frac{mg}{q} = \frac{10^{-7} \times 9.8}{1.6 \times 10^{-19}} = 6.125 \times 10^{12} \text{ N/C}$ . **Q30:**  $\tau = pE \sin \theta = (q \times 2a)E \sin 30^\circ$ .  $4 = q \times 0.02 \times 2 \times 10^5 \times 0.5 \Rightarrow q = 2 \times 10^{-3} \text{ C} = 2 \text{ mC}$ . **Q31: 1 : 3.** ( $\phi_1 = Q/\epsilon_0$ ,



$\phi_2 = (Q + 2Q)/\epsilon_0 = 3Q/\epsilon_0$ . **Q32:**  $\phi = EA \cos(0^\circ) = 5 \times 10^3 \times (0.1)^2 = 50 \text{ N} \cdot \text{m}^2/\text{C}$ .  
**Q33:**  $\vec{\tau} = \vec{p} \times \vec{E}$  or  $\tau = pE \sin \theta$ . Stable equilibrium occurs at  $\theta = 0^\circ$ . **Q34:** Draw a cylindrical Gaussian surface.  $\oint \vec{E} \cdot d\vec{A} = E \times 2\pi r l = \frac{\lambda l}{\epsilon_0} \Rightarrow E = \frac{\lambda}{2\pi\epsilon_0 r}$ . **Q35:**  $E = \frac{1}{4\pi\epsilon_0} \frac{p}{(r^2 + a^2)^{3/2}}$ .  
 Direction is opposite to the dipole moment. **Q36:**  $E = \frac{1}{4\pi\epsilon_0} \frac{2pr}{(r^2 - a^2)^2}$ . For short dipole ( $r \gg a$ ),  
 $E = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3}$ . **Q37:** Draw a pillbox Gaussian surface.  $2EA = \frac{\sigma A}{\epsilon_0} \Rightarrow E = \frac{\sigma}{2\epsilon_0}$ . **Q38:**  
 $E = \frac{1}{4\pi\epsilon_0} \frac{Qx}{(x^2 + a^2)^{3/2}}$ . **Q39:** (A) Total flux =  $q/\epsilon_0$ . (B) One face =  $q/6\epsilon_0$ . (C) At a corner, flux  
 =  $q/8\epsilon_0$ . **Q40:**  $\frac{k(1\mu)}{(x)^2} = \frac{k(4\mu)}{(30-x)^2} \Rightarrow \frac{1}{x} = \frac{2}{30-x} \Rightarrow 30 - x = 2x \Rightarrow x = 10 \text{ cm}$ . **Q41:**  
 $E = \frac{\sigma}{2\epsilon_0} = \frac{10^{-15}/18\pi}{2 \times 8.85 \times 10^{-12}} = 10^{-6} \text{ N/C}$ . It does not change with distance because the field of  
 an infinite sheet is independent of  $r$ . **Q42:** Yes, exactly at the midpoint ( $a/2, 0$ ), where fields  
 from both charges are equal and opposite, cancelling each other. **Q43:** (A)  $\phi = q_{\text{enclosed}}/\epsilon_0$ . (B)  
 (i)  $E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$ . (ii)  $E = 0$ . (C) Graph is zero until  $R$ , jumps to max at  $R$ , then decays as  $1/r^2$ .  
**Q44:** (A)  $E = \frac{\lambda}{2\pi\epsilon_0 r}$ . (B) Centripetal force = Electrostatic force  $\Rightarrow \frac{mv^2}{r} = eE = \frac{e\lambda}{2\pi\epsilon_0 r}$ . Kinetic  
 Energy =  $\frac{1}{2}mv^2 = \frac{e\lambda}{4\pi\epsilon_0}$ . **Q45:** (A)  
 $\phi = \phi_R + \phi_L = EA + EA = 2 \times 200 \times (\pi \times 0.05^2) = 3.14 \text{ N} \cdot \text{m}^2/\text{C}$ . (B)  
 $Q = \phi\epsilon_0 = 3.14 \times 8.85 \times 10^{-12} = 2.78 \times 10^{-11} \text{ C}$ . **Q46:** (A) Flux is the number of field  
 lines crossing an area; SI unit  $\text{N} \cdot \text{m}^2/\text{C}$ . (B) (i)  $\phi = E_{x_2}A - E_{x_1}A$ . (ii) Calculate  
 $Q_{\text{enclosed}} = \phi\epsilon_0$ . **Q47:** (A) Geometric vector addition of  $\vec{E}_{+q}$  and  $\vec{E}_{-q}$  components. (B) Same  
 as Q21: Middle point (1 m),  $Q = -q/4$ . **Q48:** (A)  $\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1q_2}{r^2} \hat{r}_{12}$ . (B) Vector sum of forces  
 from  $-4q$  and  $2q$  on  $q$  using  $F = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos 120^\circ}$ . **Q49:** (A)  $\vec{\tau} = \vec{p} \times \vec{E}$ . (B)  
 $\tau = (q \cdot 2a)E \sin \theta \Rightarrow 4\sqrt{3} = (8 \times 10^{-9} \times 0.04) \times E \times \sin 60^\circ \Rightarrow E = 1.44 \times 10^{10} \text{ N/C}$ .  
**Q50:** (A) Superposition of fields  $E = \sigma/2\epsilon_0$  from each plate. (B) Outer regions: fields are  
 opposite and cancel ( $E = 0$ ). Inner region: fields are in the same direction and add up ( $E = \sigma/\epsilon_0$ ).